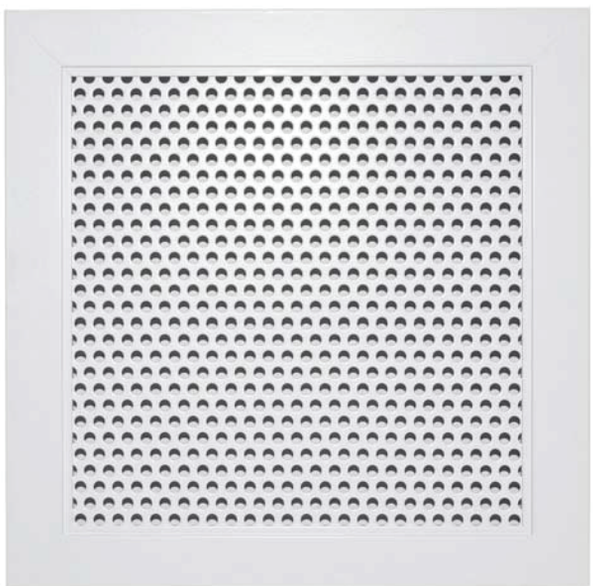
Perforated Diffuser PFF/PFDD



PFF

Description
Rectangular ceiling diffuser for exhaust/ supply of the air flow provided with circular perforations and mounting frame PFF/PFDD. PFDD is provided with access door to ease access to the back of the diffuser.

Application
Exhaust/ supply of the air flow.

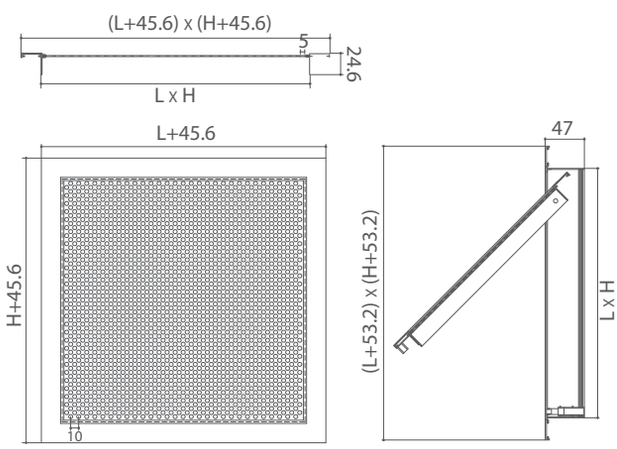
Characteristics
The diffusers are executed with mounting holes on the frame, except those with dimensions 595 x 595 mm for paneled ceiling.
PFDD is provided with access door, in this case the mounting holes will not be executed on the frame, but on the inlet connection of the diffuser.
The access door is opened via a click type system (open/ close by pressing).
Circular perforations of 5 mm diameter.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.
An air filter FIL can be optionally mounted on the inlet connection of the diffuser .
The diffuser connects to the ventilation duct through the adapter.

Accessories
Air filter **FIL**
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Galvanized steel sheet painted in electrostatic field (white RAL 9016).
Other RAL colors available on request.

Technical design

PFF **PFDD**



Assembly Method
With screws (standard).
PFDD - screws on the inlet connection (optional).

Selection table PFF/PFDD

Supply

Air flow m³/h	LxH	300x300	400x400	500x500	595x595
	Ak [m²]	0.030	0.060	0.100	0.150
160	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.5 0.9 22.0 9.0			
200	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.8 1.1 31.0 14.0	0.9 0.8 < 20 4.0		
300	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.8 1.7 44.0 31.0	1.4 1.2 25.0 8.0		
400	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	3.7 2.1 53.0 53.0	1.9 1.6 35.0 15.0	1.1 1.3 20.0 5.0	
500	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	4.6 2.5 61.0 81.0	2.3 2.0 42.0 23.0	1.4 1.6 27.0 8.0	
700	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]		3.2 2.2 52.0 45.0	1.9 2.2 39.0 16.0	1.3 1.8 27.0 7.0
1000	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]		4.6 3.1 63.0 90.0	2.8 2.4 51.0 30.0	1.9 2.6 39.0 15.0
1600	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]				3.0 3.1 54.0 36.0

Exhaust

Air flow m³/h	LxH	300x300	400x400	500x500	595x595
	Ak [m²]	0.030	0.060	0.100	0.150
160	Veff(m/s) NR [dB(A)] ΔPt [pa]	1.5 <20 20.0			
200	Veff(m/s) NR [dB(A)] ΔPt [pa]	1.8 <20 31.0			
300	Veff(m/s) NR [dB(A)] ΔPt [pa]	2.8 26.0 76.0			
400	Veff(m/s) NR [dB(A)] ΔPt [pa]	3.7 35.0 125.0	1.9 <20 33.0		
500	Veff(m/s) NR [dB(A)] ΔPt [pa]	4.6 42.0 185.0	2.3 23.0 47.0		
700	Veff(m/s) NR [dB(A)] ΔPt [pa]		3.2 34.0 95.0	1.9 20.0 37.0	
1000	Veff(m/s) NR [dB(A)] ΔPt [pa]			2.8 31.0 73.0	1.9 20.0 33.0
1600	Veff(m/s) NR [dB(A)] ΔPt [pa]				3.0 36.0 85.0

NR < 35 | 35≤NR≤45 | NR>45

Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* Vk = 0.2 m/s - Velocity at which the air jet length was calculated "x" [m]
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type	+	FIL	+	OBD	+	ADP IZ/N	/	Dimensions	/	L x H	/	0
PFF / PFDD		Air filter		Air control damper		Insulated/Non-insulated adapter		L X H 300 x 300 400 x 400 500 x 500 595 x 595		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints